Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 19)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 1 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 1 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 1 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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0 1 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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0 2 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

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Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 2 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 3 Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km^2	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km^2	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km^2	under 250	over 50% in 10 years or 3 generations

0 3 . 1 State **three** roles of the IUCN.

[3 marks]

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0 3 . 2 A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km^2 of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 3 . 3 Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 3 . 4 Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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0 4 . 1 Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 4 . 2 Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 4 . 3 Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 4 . 4 Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

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0 5 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

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Square-mesh escape panels .....

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Minimum and maximum catchable sizes .....

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No-take zones .....

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Closed seasons .....

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0 5 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 6 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

0 6 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

0 6 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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0 7 Table 1 compares organic and inorganic fertilisers.

0 7 . 1 Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0 7 . 2 Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

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0 8 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 8 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 8 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 9 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 9 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

0 9 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

0 9 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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0 9 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

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1 0 . 1 Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

Extra space

1 0 . 2 State what is meant by a **taxon**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS